

Sl.No	Question	Marks	CO
Module 1			
1.	Define memory mapped IO and IO mapped IO with examples.	8	2
2.	What are the different methods of DMA? Explain them in brief. Explain the registers in DMA	10	2
3.	What is an interrupt? with example illustrate the concept of interrupts.	8	2
4.	Describe how a read operation is performed on a PCI bus	8	2
5.	List the sequence of events that takes place when a processor sends a command to the SCSI controller	8	2
6.	Explain the following with respect to USB, I) USB architecture, II) USB addressing, III) USB protocols	10	2
7.	Define the terms 'cycle stealing' and 'block mode'	6	2
8.	What is bus arbitration ? Explain the different approaches to bus arbitration.	10	2
9.	With a timing diagram explain centralized bus arbitration/ daisy chaining technique	10	2
10.	With a timing diagram explain synchronous bus system	8	2
11.	With a timing diagram explain asynchronous bus system	8	2
12.	With neat block diagram explain standard I/O interface	8	2
13.	Briefly discuss the main phases involved in the operation of SCSI bus.	10	2
14.	Explain the tree structure of USB with split bus operation	6	2
15.	Explain the use of PCI bus in a computer with necessary figure.	10	2

16.	List the SCSI bus signals with their functions	6	2
17.	With neat diagram explain MAR, MDR, PC and IR the connection Between the Processor and the Memory	8	1
18.	Explain the USB data packets and control packet concepts with supporting diagrams.	10	2
19.	Illustrate the program controlled I/O with the program that reads one line from keyboard and echoes it the display	10	1
20.	Describe the connection between the processor and memory and List the steps needed to execute ADD LOCA,R0	8	1
21.	Explain single and Mutiple bus system with a neat diagram.	8	1
Module 2			
1.	Explain the following types of memory a. RAMBUS b. EPROM c. ROM d. Flash Drives e. CACHE memory	10	2
2.	With neat flow diagram explain Single-bus organization of the datapath inside a processor.	10	2
3.	Explain the process of fetching a word from memory considering MDR	8	2
4.	Considering Add (R2), R0 – explain the complete execution steps and control sequence	10	2
5.	Describe the control sequence for a unconditional branch instructions	8	2
6.	What is the advantage of multiple bus organization? Explain the three-bus organization of the datapath.	10	2
7.	Briefly explain the following concepts in context of hard-wired control: a. Generation of end control signal b. Separation of decoding and encoding functions	10	2
8.	Explain briefly basic organisation of a microprogrammed control unit	10	2
9.	Write and explain Add (Rsrc)+, Rdst microinstructions	8	2

10.	Illustrate how Read and write operations are performed in Static RAM cell	8	2
11.	How is CMOS memory cell realized-explain	8	2
12.	Explain the operation of Synchronous DRAM with supporting block diagram and timing diagram considering burst read.	10	2
13.	Enumerate the importance of Memory controller	8	2
14.	Differentiate Associative-mapped and set associative mapped Cache	10	2
15.	Emphasise on 2MX32 memory module using 512L X 8static memory chips	10	2
16.	Explain fast page mode ,latency and bandwidth	6	2
17.	How is memory interfaced to processor	8	2
Module 3			
1.	Differentiate CISC and RISC architectures.	6	1
2.	Explain the important design rules of RISC philosophy	6	1
3.	Which are the different features of ARM instruction set that make it suitable for embedded applications.	10	3
4.	With a neat diagram explain the different hardware components of an embedded device based on ARM core.	10	3
5.	Explain the AMBA bus protocol.	10	3
6.	Give a detailed account of different types of memories that can be used in embedded systems based on the hierarchy, width and types.	8	3
7.	What are memory controllers and interrupt controllers?	10	3
8.	With a neat diagram explain the different software components of an embedded system.	10	3

9.	Give different applications of ARM processors.	8	3
10.	Give the significance of initialization code in embedded system software.	8	3
11.	With a neat diagram explain the different general purpose registers of ARM processors.	10	3
12.	Explain ARM core dataflow model with a neat diagram.	10	3
13.	Differentiate ARM and Thumb instruction set features.	10	3
14.	Explain current program status register (CPSR) with neat diagram.	10	3
15.	What are banked registers? Show how the banked registers are utilized when the user mode changes to IRQ mode	10	3
16.	Which are the different conditional flags of ARM processor.	10	3
17.	Explain ARM pipeline with 3,5,6 stages.	10	3
18.	What are interrupts or exceptions? How are they handled in ARM processors?	8	3
19.	Explain Non-protected memory, MPU & MMU.	8	3
20.	What are the different techniques of core extensions?	8	3
Module 4			
1.	Explain different data processing instructions in ARM	10	3
2.	Explain the MOV instruction set provided by ARM with the example for each	8	3
3.	Explain the barrel shifter operation in ARM processor, with neat diagram	6	3
4.	Briefly explain the different Load- Store Instruction categories used with ARM	10	3
5.	Write a program for forward and backward branch by considering an example	6	3

6.	Explain the Co-processor instruction of ARM processor	6	3
7.	Show the post condition when MOVS instruction shifts register r1 left by one bit and result is stored in r0. Where r0 = 0x00000000, r1 = 0x80000004 and CPSR= nzcvqiFt	4	3
8.	Discuss the Load Store instruction with respect to: (i) Single register transfer (ii) Multiple register transfer.	8	3
9.	What are the various logical instructions supported by ARM? Explain with example for each.	6	3
10.	Illustrate the advantage of conditional execution with c code and compare with assembly language	8	3
11.	Explain the following instructions with syntax and example. (i) MOV (ii) SWI (iii) MRC (iv) LDRB V) QADD	10	3
12.	Write ARM assembly program to add two 32-bit numbers.	8	3
13.	Explain the following with examples. (i) RSC (ii) MLA (iii) STRH (iv) SWP v) BIC VI) TST	10	3
14.	Explain the multiply instructions of ARM processor	6	3
15.	Explain stack operation with an example	6	3
16.	Write an ALP to add an array of 16-bit numbers and store the 32-bit result in internal RAM.	8	3
17.	Write ARM assembly program to find the sum of first 10 numbers	4	3
18.	Write ALP to add and multiply 2 numbers and show results through registers	6	3
19.	Write ALP to find sum of integers between 0 to 9 or 1 to 10	6	3
20.	Write ALP to find factorial of a given number	8	3
21.	Write ALP to find square and cube of a given number	8	3
22.	Write ALP to add first 5 even numbers(0 to 8) or Odd numbers(1 to 9)	8	3

Module 5			
1.	Explain ARM Thumb interworking with an example.	10	4
2.	Explain the following instructions with example. i) MOV with v1 ii) BLX with v2 iii) TST with v1 iv) EOR with v1 V) SUB with v1	10	4
3.	Differentiate between ARM and Thumb instructions	6	4
4.	Explain the operation of stack instruction in thumb mode	6	4
5.	Justify how code density increases in thumb instruction	6	4
6.	Write an ALP to add two 16 bit numbers using thumb instructions.	6	4
7.	Explain signed vs unsigned data types used in C programming	6	4
8.	Explain loop unrolling with an example.	10	4
9.	Write a C function to find check sum.	10	4
10.	Write Advantages and disadvantages of C programming.	6	4
11.	Differentiate between C and ALP	6	4
12.	Explain data processing thumb instructions with examples.	10	4
13.	Illustrate PUSH and POP instruction with subroutine example.	10	4
14.	Explain the process of handling the interrupts in THUMB state	10	4
15.	Write ALP to transfer data from one location to another using internal RAM	8	4
16.	Write ALP to find square of a given number using look up table	5	4

17.	Write an ALP to count the number of ones and zeros in two consecutive memory locations.	6	4
18.	Interface a Stepper motor and rotate it in clockwise and anti-clockwise direction	10	4
19.	Interface a DAC and generate Triangular and Square waveforms.	8	4
20.	Interface a DC motor to and write ALP to control its speed.	8	4